

MADIS ALTMANN

Senior Python Full-Stack Engineer

FastAPI, Django, React, AWS, Microservices, AI/ML

CONTACTS

Narva, Estonia | Open to Remote / Relocation

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EDUCATION

M.S., Computer Science

University of Tartu

Sep 2010 - Jun 2015

GPA: 4.6/5.0

COMPETENCIES

- Backend Engineering (Python)
- Full-Stack Development
- Distributed Systems & Microservices
- AI-Driven Product Development
- Real-Time & Event-Driven Systems
- Database & Data Engineering
- Asynchronous Processing & Performance Optimization
- Cloud & DevOps (AWS/GCP)
- Scalable Cloud Infrastructure
- System Design & Architecture
- Agile Development & Collaboration

ACHIEVEMENTS

- Reduced manual documentation by 70% by delivering real-time AI transcription and summarization.
- Scaled distributed systems to support hundreds of concurrent real-time sessions with low latency and high reliability.
- Achieved millisecond-level search performance through advanced indexing and query optimization.
- Improved system responsiveness by implementing asynchronous, non-blocking processing pipelines.
- Increased system availability to 99.9%+ uptime with cloud-native, auto-scaling infrastructure.
- Boosted user engagement by 30% through AI-powered voice and conversational systems.
- Reduced operational workload by 75% via automation of core business workflows.
- Enabled real-time data processing using event-driven architecture and messaging systems.

SUMMARY

Senior Python Software Engineer with 10+ years of experience building scalable backend systems, AI-powered platforms, and cloud-native applications across healthcare, media, and enterprise communication domains. Strong expertise in Python, FastAPI, Django, microservices, distributed systems, and AWS-based architectures, with hands-on experience integrating LLM and NLP solutions using OpenAI, LangChain, and RAG pipelines. Experienced in developing high-performance APIs, real-time systems, and production-grade cloud infrastructure within Agile environments.

CORE TECHNOLOGIES

Python	FastAPI	Django	REST APIs	Microservices	PostgreSQL
Redis	Kafka	Docker	Kubernetes	AWS	React
TypeScript	OpenAI	LangChain	RAG	CI/CD	GitHub Actions
Linux					

WORK EXPERIENCE

Senior Full-Stack Software Engineer

Apr 2023 – Mar 2026

Innowise | Paris, France

SVENSA Voice – AI-powered conferencing platform

- Led backend development for AI-powered real-time conferencing using **Python 3.11–3.12**, **Django 4.2–5.0**, **FastAPI**, **AWS**, and **LiveKit**, enabling multilingual transcription and meeting automation.
- Designed scalable microservices and low-latency pipelines using **REST APIs**, **gRPC**, asynchronous processing and distributed system principles.
- Built high-performance data platforms with **PostgreSQL 14–16**, **SQLAlchemy**, optimizing query performance for transcripts and analytics.
- Developed **async task pipelines** with **Redis 7** and **Celery 5.3+**, improving real-time AI inference and workflow reliability.
- Managed cloud-native infrastructure using **Docker**, **Kubernetes (EKS 1.26+)**, **AWS ECR**, **CloudWatch**, and **GitHub Actions CI/CD** for scalable and reliable deployments.
- Integrated **GenAI** capabilities using **OpenAI GPT-4/4o**, **Claude 3**, **LangChain**, and **RAG pipelines** for meeting summarization and conversational intelligence.
- Developed frontend applications using **React 18**, **TypeScript**, **Next.js**, and **D3.js** with **SSR/SSG**, responsive UI, and accessibility best practices.

Senior Software Engineer

Aug 2020 – Mar 2023

ELEKS | Tallinn, Estonia

MemoryVoice – AI Voice Assistant (Healthcare / Digital Health)

- Built secure backend services for AI-driven healthcare using **Python 3.9–3.10**, **FastAPI**, **AWS S3**, and **IAM**, supporting voice-based patient interactions.
- Architected microservices and **async systems** with **REST APIs**, **TDD** and **SOLID** principles for conversational AI and contextual memory.
- Designed **data/search solutions** with **MongoDB 4.4–6.0**, **DynamoDB**, **Elasticsearch 7**, improving patient history retrieval performance.
- Implemented low-latency pipelines using **Redis 6**, **Celery 5.2+**, and caching, enhancing real-time voice response.
- Deployed infrastructure on **Docker**, **AWS EKS (1.21–1.24)**, **Lambda** with encryption and scalable, compliant architectures.
- Built **AI features** with **NLP**, **Speech-to-Text**, **Text-to-Speech**, **Pandas**, **NumPy**, **SciPy**, and **PySpark** for intelligent patient engagement.
- Developed frontend with **Angular**, **RxJS**, and **Reactive Forms**, focusing on accessibility, **HTML5**, and mobile performance.

Software Engineer

Jan 2019 – Jul 2020

Chudovo OÜ | Narva, Estonia

AI Voice Concierge – Hospitality & Travel Platform (SaaS / Automation)

- Developed backend systems using **Python 3.7–3.8**, **Django 2.2–3.1**, and **AWS Route53** for hospitality automation and operational workflows.
- Implemented **API** integrations with **FastAPI**, **REST APIs** and **PMS/CRM/POS** systems.
- Optimized databases using **MySQL 5.7/8.0**, **PostgreSQL 11–12** with advanced indexing and query tuning.
- Built **event-driven pipelines** with **Kafka 2.3–2.6** and **AWS Lambda** for real-time notifications and automation.
- Managed infrastructure on **AWS EKS (1.17–1.18)**, **S3** with high-availability designs.
- Implemented **NLP-based** voice automation for multilingual guest interactions.
- Built frontend with **Vue.js**, focusing on responsive UI and performance optimization.

Full-Stack Engineer

Jul 2015 – Dec 2018

Ronas IT | Tallinn, Estonia

MediaSphere – AI Content Personalization Platform (Media / EdTech)

- Built backend systems using **Python 3.5–3.7**, **Django 1.11–2.1** on **GCP** for media distribution across web, mobile and smart TV.
- Designed services with **GraphQL**, **REST APIs**, **WebSockets** and **CQRS**-based patterns.
- Developed data layer with **MongoDB 3.4–4.0**, **DynamoDB** for content and recommendation systems.
- Implemented streaming pipelines with **Kafka 1.4–2.0** and **Redis pub/sub**.
- Managed global delivery using **CloudFront**, **IAM** and multi-region architectures.
- Built AI-driven personalization features improving user engagement and retention.
- Developed frontend with **React 16** and **Next.js 7–8** using **SSR/SSG** for performance.